Allan Paiz

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Education

- B.S. in Computer Science, University of South Carolina — Expected Dec 2026
  Minor in Mathematics | GPA: 3.7/4.0

Coursework: Software Engineering, Analysis of Experimental Data, Optimization, Linear Algebra, Math Foundations of ML/DS, AI, Computer Architecture, Operating Systems, Technical Writing, Elementary Number Theory, Computer Networks

Skills

Programming: Python, C++, Java, SQL, Bash; pandas, NumPy, scikit-learn, TensorFlow, Matplotlib, JavaFX, JUnit, SQLite
Tools: Linux, LaTeX, Excel, Git, GitHub, ROS 2, Docker, Gazebo, Unreal Engine 5
AI-Assistance: ChatGPT, OpenAI Codex, Claude; agent-assisted prototyping, debugging, code review, and testing
Spoken Languages: Spanish (native), French (elementary), Russian (elementary)

Experience

Undergraduate Research Assistant, University of South Carolina – Columbia, SC — *Jan 2026 – Present*

- Develop and debug Python and C++ robotics software on Linux for autonomous navigation and multiagent path finding, supporting simulation experiments evaluating coordination across 8 to 128 agents.
- Conduct ROS 2 hardware experiments with five TurtleBot4 robots, collecting trajectory and communication data for a multiagent path planning evaluation that achieved 100% task success across five scenarios.
- Implement telemetry logging and data visualization for robot trajectories and network communications, supporting research that demonstrated 2 to 510 times lower information sharing than baseline planning algorithms in simulation.
- Support hardware evaluation of event triggered optimal control on TurtleBot4 robots, collecting experimental data for research reporting 65% fewer control updates and 35.7 mm tracking RMSE across four scenarios.
- Integrate LiDAR and camera sensing into Docker based Gazebo simulations of ground and aerial robot fleets to investigate robotic perception, autonomous navigation, and multirobot coordination.

Researcher, Dynamical Systems & AI Research Group – Columbia, SC — *May 2026 – Aug 2026*
Research Experience for Undergraduates — ***Links:*** Perception Report

- Developed a Gazebo simulation of multiple robots using Python and Bash, leveraging LLMs and AI coding agents to accelerate learning, prototyping, debugging, code review, and iterative testing.
- Implemented YOLO based object detection and LiDAR mapping to investigate computer vision, robotic perception, and autonomous navigation.
- Designed and documented a ROS 2 perception architecture incorporating OAK stereo vision, 2D LiDAR, object tracking, sensor fusion, and safety aware control to support autonomous robot navigation.

Research Presentation

Collaborative Autonomy in Uncertain Environments: Exploring Vistas Beyond Consensus – Panama City, FL — *Oct 2026*
Naval Engineering Education Consortium (NEEC)

- Co-presented Navy funded collaborative autonomy research on multiagent path finding, explainable AI, and resource efficient control, communicating robotic coordination methods and experimental findings to an engineering audience.
- Presented contributions in LiDAR sensing, camera perception, Gazebo simulation, and robot telemetry collection, connecting robotics software development to experimental validation and autonomous systems research.

Projects

Event-Triggered Optimal Control for Obstacle-Avoidant Navigation — ***Links:*** Course Paper

- Developed Python simulations to analyze autonomous robot navigation under motion constraints and obstacle avoidance requirements, documenting optimization methods and results in a technical paper.
- Formulated trajectory planning for nonholonomic robots as a constrained nonlinear optimization problem, incorporating robot motion limitations to evaluate feasible navigation paths.
- Investigated event triggered control strategies for autonomous navigation, evaluating tradeoffs among collision avoidance, trajectory feasibility, and control effort.

Construction Permit Delay Risk Prediction — ***Links:*** Project Paper Portfolio

- Analyzed 7,504 Atlanta residential building permits to identify projects at risk of approval delays, applying predictive

modeling to support construction bid planning and decision making.
- Preprocessed structured permit data using Python, pandas, NumPy, and SQLite, engineering features from location, zoning, land use, and application dates for machine learning classification.
- Benchmarked kernel SVM and neural network classifiers on structured permit data, achieving 69.13% classification accuracy with the neural network compared to 48.35% with the SVM.

Defensive Stopping Power – 2024 NFL Big Data Bowl Finalist **Links:** GitHub Kaggle Submission

- Earned finalist recognition in the 2024 NFL Big Data Bowl by developing a Python based analysis of NFL player tracking data to quantify individual defenders' tackling impact.
- Engineered spatial, velocity, and play context features from NFL tracking data to train a feedforward neural network predicting each defender's probability of contacting the ball carrier.
- Developed the Yards Allowed After Contact (YAAC) metric using inferred contact events to quantify defensive performance and support player rankings and scouting analysis.

Green Dog – 2025 NFL Big Data Bowl **Links:** GitHub Kaggle Submission

- Developed a supervised machine learning classification model using NFL tracking data to predict whether eligible receivers would run routes or remain in pass protection before the snap.
- Designed a three branch neural network using TensorFlow and Keras, combining engineered feature groups to estimate receiver route probabilities.
- Evaluated model performance on subsequent game weeks using accuracy, precision, recall, F1 score, and ROC AUC, preserving chronological ordering to reduce data leakage during model validation.

Wash Your Rice – Unreal Engine 5 Capstone **Links:** Capstone Final Demo

- Collaborated with a five person development team to design, implement, and release a horror puzzle game using C++ and Unreal Engine 5.
- Developed an interactive computer interface featuring dynamic screens, minigames, and mouse and keyboard input to support interconnected gameplay and puzzle mechanics.
- Implemented automated software tests for interface states and puzzle behavior to validate functionality and support integration across gameplay systems.

TuneUp by JETA – Java Music Application **Links:** GitHub Development Demo

- Collaborated with a four person software development team to build a Java music application supporting song libraries, audio playback, and interactive free play modes.
- Translated client requirements into software specifications and UML diagrams, applying object oriented programming principles and design patterns to guide application development.
- Implemented and unit tested Java application features using JUnit and JSON backed data, coordinating development through Agile Scrum sprints with Git and GitHub Projects.

Other Work History

Advanced Data Collector, Pro Football Focus – Remote *May 2024 – Present*

- Collect structured NFL and NCAA player and play level data, contributing to datasets used in Pro Football Focus analytics and client products.
- Perform data quality assurance by reviewing and validating game records, identifying discrepancies, and correcting errors to maintain consistency and accuracy across datasets.
- Deliver accurate game data under recurring reporting deadlines throughout the football season, balancing data collection volume, quality, and timeliness.

Combat Engineer, United States Marine Corps – California *May 2018 – Oct 2022*

Maintained a Secret security clearance

- Led small teams during field engineering operations, coordinating personnel and task execution in mission critical environments.
- Utilized the Engineer Field Reconnaissance GIS system to collect, map, and analyze terrain and route information supporting operational planning.
- Instructed junior Marine leaders at Sapper Leaders Course on planning engineering tasks, coordinating personnel and equipment, and executing field operations.